Model Validation Report

1. Executive Summary

Overall risk is MEDIUM. Final decision is insufficient_evidence. Primary reasons: Missing monitoring plan rule triggered., Missing documentation rule triggered.. Insufficient evidence items: candidate model artifact, model documentation, user-supplied dataset.

2. Model Overview

Target column: defaulted. Problem type: classification.

3. Files Reviewed

dataset: sample_credit_risk.csv, dataset: generated sample credit risk dataset

4. Validation Scope

Scope included data quality, methodology, performance, explainability, fairness, governance, and final decisioning.

5. Data Quality Review

Dataset has 500 rows and 11 columns.

6. Methodology Review

Methodology review considered available model artifact, training code, baseline, and reproducibility evidence.

7. Performance Review

Performance review interprets Python-calculated metrics only.

8. Explainability Review

Explainability review is based on generated Python artifacts.

9. Bias and Fairness Review

Fairness review used available group attributes and Python-calculated group metrics.

10. Governance Review

Governance review considered owner, versioning, monitoring, retraining, lineage, and oversight evidence.

11. Key Findings

- **MEDIUM | Documentation: Model documentation not supplied**. Evidence: No documentation file was uploaded. Recommendation: Provide model purpose, scope, data description, methodology, limitations, monitoring, governance, and approval evidence.

- **MEDIUM | Data Quality: Outliers detected in numeric features**. Evidence: 5 numeric columns contain IQR outliers. Recommendation: Review outlier treatment and confirm values are plausible.

- **MEDIUM | Data Quality: Target class imbalance detected**. Evidence: Minority class rate is 0.164. Recommendation: Use imbalance-aware metrics, sampling, or class weights and monitor minority-class performance.

- **MEDIUM | Methodology: Training code not supplied**. Evidence: The validation package does not include training code. Recommendation: Provide reproducible training code with feature engineering, split logic, hyperparameters, and model artifact creation.

- **LOW | Methodology: Independent validation used a baseline model**. Evidence: No trained model artifact was supplied, so deterministic analysis trained a baseline model. Recommendation: Provide the candidate model artifact for direct validation before production approval.

- **LOW | Performance: Performance evidence is acceptable for baseline validation**. Evidence: Deterministic test metrics were calculated for classification. Recommendation: Compare these results with business thresholds and production challenger models.

- **LOW | Explainability: Explainability artifacts generated**. Evidence: Feature importance and/or SHAP plots were generated by Python tools. Recommendation: Review top drivers for business plausibility and prohibited proxies.

- **MEDIUM | Fairness: Potential group performance disparity detected**. Evidence: Disparities: [{'column': 'gender', 'metric': 'accuracy', 'max_gap': 0.04578246392896779}, {'column': 'region', 'metric': 'accuracy', 'max_gap': 0.21734892787524362}]. Recommendation: Investigate group-level errors, thresholds, and mitigation controls before approval.

- **MEDIUM | Governance: Monitoring plan is missing or weak**. Evidence: Monitoring, drift, or retraining triggers were not identified in the supplied evidence. Recommendation: Define monitoring metrics, thresholds, ownership, escalation, and retraining triggers.

- **MEDIUM | Governance: Approval evidence is missing**. Evidence: Formal approval evidence was not identified. Recommendation: Document model owner, version, approvers, approval date, and production-use conditions.

12. Challenge Questions

- What business decision does the model support and where is this documented?

- What assumptions and limitations were accepted by model governance?

- Which upstream controls prevent target-derived fields from entering production scoring?

- How are missing values and outliers treated consistently between training and production?

- Why was this algorithm selected over simpler challenger models?

- How was the train/test split designed to avoid temporal or entity leakage?

- What minimum performance threshold is required for the business use case?

- How stable are these metrics across time, segments, and out-of-sample cohorts?

- Do the top model drivers align with business expectations?

- Could any high-importance feature act as a proxy for a protected class?

- Which protected-class or proxy attributes are approved for fairness testing?

- What disparity thresholds trigger remediation or governance escalation?

- Who owns model performance and issue remediation after deployment?

- What thresholds trigger model retirement, recalibration, or retraining?

13. Risk Rating

MEDIUM

14. Recommendations

- Maintain a complete model document with owner, version, scope, monitoring, and approvals.

- Review outlier treatment and confirm values are plausible.

- Use imbalance-aware metrics, sampling, or class weights and monitor minority-class performance.

- Provide reproducible training code with feature engineering, split logic, hyperparameters, and model artifact creation.

- Provide the candidate model artifact for direct validation before production approval.

- Compare these results with business thresholds and production challenger models.

- Review top drivers for business plausibility and prohibited proxies.

- Investigate group-level errors, thresholds, and mitigation controls before approval.

- Define monitoring metrics, thresholds, ownership, escalation, and retraining triggers.

- Document model owner, version, approvers, approval date, and production-use conditions.

15. Final Validation Decision

insufficient_evidence

Deterministic Artifacts

- confusion_matrix: `figures/confusion_matrix.png`

- feature_importance: `figures/feature_importance.png`

- shap_summary: `figures/shap_summary.png`